

The Henry and Marilyn Taub
Faculty of Computer Science

Module 1: Fundamental Concepts in Structured ML

Selected Topics on **Deep Learning over Structured Data** (2360028)

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Module Goals

- Understand the core data models of the course: *graphs*, *tables*, and *relational databases*
- See how relational databases can be analyzed using graph concepts and tools
- Develop familiarity with *key graph concepts* relevant to the course
- Review *fundamental machine learning* (ML) principles
- Introduce basic concepts in *ML for structured data*

Outline

1. Data Models
   - Graphs
   - Tables
   - Relational Databases
2. Databases as Hetero-Graphs
3. Required Graph Concepts
   - Neighborhood
   - Matrix Representations of Graphs
   - Weisfeiler-Leman Isomorphism Test
4. ML on Structured Data
   - General ML
   - Predictive Tasks
   - Training

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Database: Collection of Tables

Student

sid	name	started	address	gender	track
89	Anna	2022	Haifa	F	CS3
60	Bill	2024	Haifa	M	CS4
52	Carry	2023	Tel-Aviv	F	SE

Take

sid	cid	year	semester
89	12	2025	1
89	25	2024	2
60	25	2024	2

Course

cid	name	credits	faculty
12	DB	3	CS
25	ML	3	CS
34	OS	4	ECS

Database: Collection of Tables **with Connected Tuples**

Student

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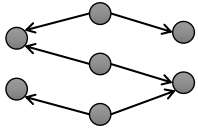
Our Data Models

- We focus on **machine learning over relational databases**
- A relational database consists of tables with *connected records* (rows)
- Hence, a natural representation of a database uses graphs
  - Where records serve as nodes
- Therefore, we will learn machine learning techniques over 3 data models: graphs, tables, and relational databases

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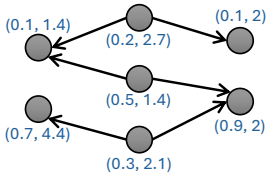
Basic Graphs



- *Directed graph*: (V, E) where V is a set of *nodes* and $E \subseteq V \times V$ is a set of *edges*
- *Undirected graph*: $E \subseteq \{\{u, v\} \subseteq V \mid u \neq v\}$

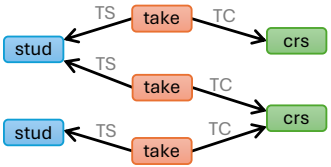
Unless stated explicitly, in this lecture we assume graphs are **directed**

Featured Graphs



- In a *featured graph*, every node is associated with a numerical *feature vector*
- Formally, a d-dimensional featured graph is a triple $(V, E, \mathbf{X})$ where
  - (V, E) is a graph
  - $\mathbf{X} : V \to \mathbb{R}^d$ associates a *feature vector* with every node
    - $\mathbf{X}$ is typically represented as a matrix in $\mathbb{R}^{|V| \times d}$, assuming a predefined ordering $v_1, v_2, \ldots$ over V
    - We often denote the vector $\mathbf{X}(v)$ by $\mathbf{x}_v$
- In rare cases, we may also discuss *edge-featured* graphs, where each edge has a feature vector
  - (of some edge-feature dimension)

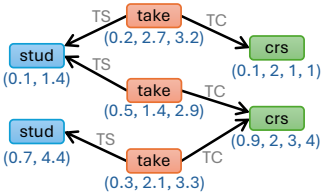
Labeled Graphs



- We often want the vertices and edges to be *labeled* (or *typed*)
- A *labeled graph* is a tuple $(L_V, L_E, V, E, \lambda)$ where:
  - L_V and L_E are sets of *node labels* and *edge labels*, respectively
  - V is the node set and $\lambda : V \to L_V$ assign a label (type) to each node
  - $E \subseteq V \times L_E \times V$ is a set of labeled edges (v_1, τ, v_2)
    - By a slight abuse of notation, we denote by $\lambda(e)$ the label τ of the edge $e = (v_1, \tau, v_2)$
- Important: we assume that edge labels are consistent with node labels: for all edges (v_1, τ, v_2) and (v_1', τ', v_2'):

$$\text{if } \tau = \tau', \text{ then } \lambda(v_1) = \lambda(v_1') \text{ and } \lambda(v_2) = \lambda(v_2')$$

Heterogeneous Graphs



- A *heterogeneous graph* (or *hetero-graph* for short) is a labeled and featured graph
- With one difference: nodes of different types may have features of different dimensions
- Formally, it is a tuple $(L_V, L_E, V, E, \lambda, d, \mathbf{X})$ where:
  - $(L_V, L_E, V, E, \lambda)$ is a labeled graph
  - $d : L_V \to \mathbb{N}$ associates a *feature dimension* with each node label
  - $\mathbf{X}$ is a function mapping each node v to a feature $\mathbf{x}_v \in \mathbb{R}^{d(\lambda(v))}$
- When we wish to stress that the graph is **not** heterogeneous, then we call it *homogeneous*
  - Note: homogeneous is the same as heterogeneous where all nodes have the same label (hence the name)

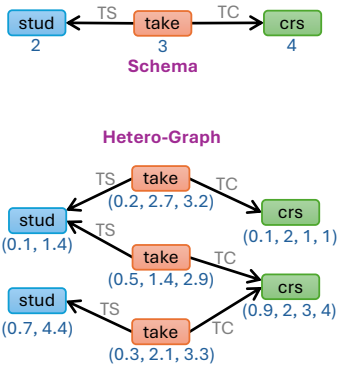
Type Consistency of Hetero-Graphs

- Two hetero-graphs $(L_V, L_E, V, E, \lambda, d, \mathbf{X})$ and $(L'_V, L'_E, V', E', \lambda', d', \mathbf{X}')$ are *type consistent* if:
  - They have the same sets of labels:

$$L_V = L'_V \quad \text{and} \quad L_E = L'_E$$

  - Every edge label is associated with the same in/out node labels in both graphs
  - $d = d'$, that is, every label $\tau \in L_V$ has the same feature dimension in both graphs
- In other words, two hetero-graphs are consistent if they have the same *schema*:

A Schema for Heterogeneous Graphs



- A *hetero-graph schema* is a tuple $(L_{\mathsf{v}}, L_{\mathsf{e}}, M, \delta)$ where:
  - L_{v} is a set of node labels
  - L_{e} is a set of edge labels
  - $M : L_{\mathsf{e}} \to L_{\mathsf{v}} \times L_{\mathsf{v}}$ maps every edge label into a pair of node labels
  - $\delta : L_{\mathsf{v}} \to \mathbb{N}$ is a dimension function
- A hetero-graph $G = (L_V, L_E, V, E, \lambda, d, \mathbf{X})$ over the schema $(L_{\mathsf{v}}, L_{\mathsf{e}}, M, \delta)$ satisfies:
  - $L_V = L_{\mathsf{v}}$ and $L_E = L_{\mathsf{e}}$
  - Every edge (u, τ, v) is such that $M(\tau) = (\lambda(u), \lambda(v))$
  - $d = \delta$
- Similar concepts: *network schema* [Samy et al., 2022]

Meta-Paths over a Hetero-Graph Schema

- We will often encounter the concept of a *meta-path* for typing *paths* (or *walks*) within a hetero-graph
- A *meta-path* Φ over the hetero-graph $(L_{\mathsf{v}}, L_{\mathsf{e}}, M, \delta)$ is a sequence of the form

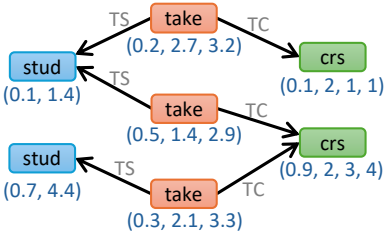
$$\sigma_0 \xrightarrow{\tau_1} \sigma_1 \xrightarrow{\tau_2} \sigma_2 \ldots \sigma_{k-1} \xrightarrow{\tau_k} \sigma_k$$

where:

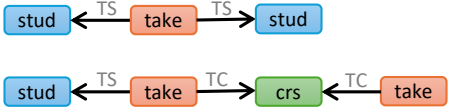
- Each σ_i is in L_{v}
- Each τ_i is in L_{e}
- $M(\tau_i) = (\sigma_{i-1}, \sigma_i)$ or $M(\tau_i) = (\sigma_i, \sigma_{i-1})$ for $i = 1, \ldots, k$

Illustration

Hetero-graph



Meta-paths



Paths/Walks over Meta-Paths

- Let $G = (L_V, L_E, V, E, \lambda, d, \mathbf{X})$ be a hetero-graph over $(L_\mathsf{v}, L_\mathsf{e}, M, \delta)$
- An undirected path $v_0, \ldots, v_k$ of G *conforms* to the meta-path Φ if for $i = 1, \ldots, k$ it holds that $\lambda(v_{i-1}) = \sigma_{i-1}$, and $\lambda(v_i) = \sigma_i$, and there is a τ_i-edge either from v_{i-1} to v_i or from v_i to v_{i-1}

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Tables

- A *tabular signature* $\mathcal{T}$ is pair $(\mathbf{A}, \mathbf{dom})$ where:
  - $\mathbf{A} = A_1, \ldots, A_n$ is a sequence of *attribute names* (or just *attributes* for short)
  - $\mathbf{dom} = dom_1, \ldots, dom_n$ is a sequence of *domains* corresponding to $A_1, \ldots, A_n$, respectively
    - Each dom_i is a finite or infinite set
    - We may write $dom(A_i)$ instead of dom_i
- A *table* (or *relation*) r over $\mathcal{T}$ is a finite subset of $dom_1 \times \cdots \times dom_n$
  - Note: we use set semantics (duplicate tuples disallowed)
- Each element $\mathbf{t} \in r$ is called a *tuple* (and sometimes a *row* or a *record*)

Domains:

[10-99]	text	[0-10]	CS, ECS, Math,...
cid	name	credits	faculty
12	DB	3	CS
25	ML	3	CS
34	OS	4	ECS

Projection

- If $\mathbf{t} = (a_1, \ldots, a_n) \in r$ is a tuple, then we may refer to each a_i as $\mathbf{t}.A_i$ or $\mathbf{t}[A_i]$
- If $\mathbf{B} = B_1, \ldots, B_\ell$ is a sequence of attributes from the signature, then we define

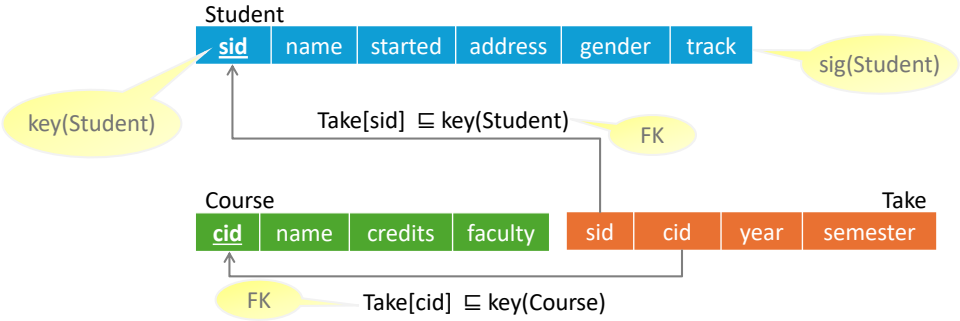
$$\mathbf{t}[\mathbf{B}] := (\mathbf{t}.B_1, \ldots, \mathbf{t}.B_\ell)$$

That is, $\mathbf{t}[\mathbf{B}]$ is the *projection* of $\mathbf{t}$ to $\mathbf{B}$

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Example of a Relational Schema



Definition of a Relational Schema

A *relational schema* $\mathcal{S}$ is a tuple $(\mathcal{R}, \mathrm{sig}, \mathrm{key}, \mathbf{FK})$ where:

- $\mathcal{R}$ is a finite set of *relation names*
- sig maps each relation name R to a tabular signature $\mathrm{sig}(R)$
- key maps every relation name R into a *primary key*, which is a sequence of attributes from $\mathrm{sig}(R)$
- $\mathbf{FK}$ is a set of *foreign-key constraints* (*FK*s for short)

Relational Schema Notation

- An FK over the schema $\mathcal{S} = (\mathcal{R}, \mathrm{sig}, \mathrm{key}, \mathbf{FK})$ has the form $R[\mathbf{C}] \sqsubseteq \mathrm{key}(S)$:
  - R and S are relation names in $\mathcal{R}$
  - $\mathbf{C}$ is a sequence of attributes from $\mathrm{sig}(R)$
  - $\mathbf{C}$ and $\mathrm{key}(S)$ have the same length
- We sometimes say that a relation R "has no key" to mean that the $\mathrm{key}(R)$ consists of all attributes
  - Which is very different from "$\mathrm{key}(R) = \emptyset$" (meaning what?)
- We denote by $dom(R)$ the domain $dom_1 \times \cdots \times dom_n$ of a relation name $R \in \mathcal{R}$ with $\mathrm{sig}(R) = (\mathbf{A}, \mathbf{dom})$ where $\mathbf{dom} = dom_1, \ldots, dom_n$

Databases

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- Let $\mathcal{S} = (\mathcal{R}, \text{sig}, \text{key}, \mathbf{FK})$ be a database schema
- A *database* D over $\mathcal{S}$ maps every $R \in \mathcal{R}$ to a table $D(R)$ over $\text{sig}(R)$ so that:
  - Primary keys are respected: for every relation name R, no two distinct tuples $\mathbf{t}_1, \mathbf{t}_2 \in D(R)$ satisfy $\mathbf{t}_1[\text{key}(R)] = \mathbf{t}_2[\text{key}(R)]$
  - FKs are respected: for every $R[\mathbf{C}] \sqsubseteq \text{key}(S)$ in $\mathbf{FK}$ and tuple $\mathbf{t} \in D(R)$ there is a tuple $\mathbf{s} \in D(S)$ such that $\mathbf{t}[\mathbf{C}] = \mathbf{s}[\text{key}(S)]$

Database "Facts"

- If D is a database over $\mathcal{S}$ and $\mathbf{t} \in D(R)$ for some $R \in \mathcal{R}$, then we call the expression $R(\mathbf{t})$ a *fact* (of D)
  - For example, $\text{Take}(89, 12, 2025, 1)$ is a fact
- We may identify a database as *the set of its facts*
  - For example, $D_1 \subseteq D_2$ means that every relation of D_1 is contained in the corresponding relation of D_2

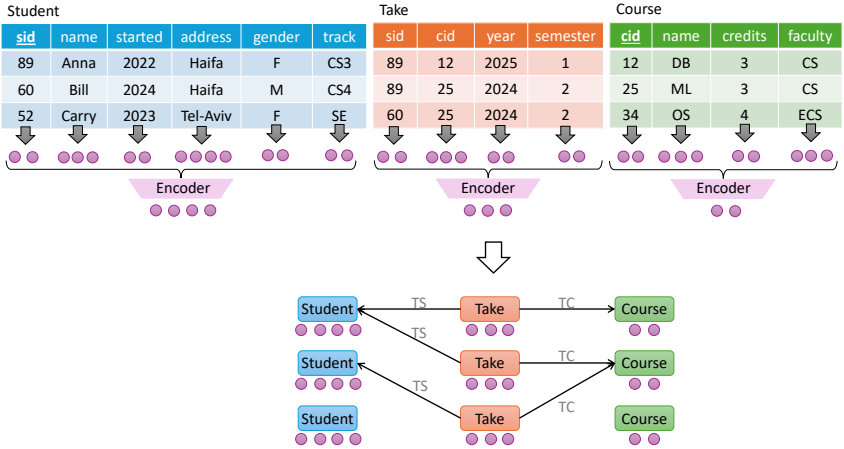
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Databases as Heterogeneous Graphs

- Various ways have been proposed for translating relational databases into graphs
- We follow a common conversion
  - E.g., [Robinson et al., 2024] [Wang et al., 2025] [Fey et al., 2024]
  - Sometimes referred to as *Row2Node* [Wang et al., 2024]

Illustration



Value Encoding

- The conversion (like many ML models over tables/databases) requires a representation of each tuple as a numerical vector
- To this aim, we first encode each cell as a numerical vector
- Cell encoding is typically determined by the modality (i.e., the type) of the domain of the cell value
  - E.g., text, numbers, categories, timestamps, images
- Off-the-shelf encoders provided in ML libraries
  - E.g., pytorch frame[1] and scikit-learn preprocessing[2]

[1] https://pytorch-frame.readthedocs.io/en/latest/index.html
[2] https://scikit-learn.org/stable/api/sklearn.preprocessing.html

Examples of Encodings

- A *categorical* value can be encoded by numbering categories, by one-hot encoding, by embedding each category into a learned feature vector, etc.

sid	degree-level	cat#	one-hot	embed
89	UG	1	(1,0,0)	(0.41,0.2,0.35)
60	masters	2	(0,1,0)	(0.98,0.55,0.59)
26	UG	1	(1,0,0)	(0.41,0.2,0.35)
52	doctorate	3	(0,0,1)	(0.88,0.42,0.4)
56	masters	2	(0,1,0)	(0.98,0.55,0.59)

- A *numerical* value x can undergo a scaling w.r.t. the dataset (e.g., to obtain numbers in $[0, 1]$), or a linear encoding $(w_1 x, w_2 x, \dots)$ where the w_i are weights
  - Predefined weights or learned as part of training

Examples of Encodings (cont'd)

- A *textual* value can be assigned an embedding from a text model
  - E.g., a transformer that is either *fully precomputed* or with *learned parameters*
- A *timestamp* value can be encoded simply via the Epoch converter, or (more commonly) as an encoding of the time's individual features, e.g., month, day, day-of-the-week, hour, minute, using direct or cyclic encoding

Timestamp (UTC)	Epoch (sec)	Extracted components	Circular (hour)
2026-01-09 00:00:00	1767916800	h = 0, d = 9, m = 1, wd=6	$(\sin 0,\ \cos 0)$
2026-01-09 06:00:00	1767938400	h = 6, d = 9, m = 1, wd=6	$\left(\sin \frac{\pi}{2},\ \cos \frac{\pi}{2}\right)$
2026-01-09 18:00:00	1767981600	hour = 18, d= 9, m = 1, wd=6	$\left(\sin \frac{3\pi}{2},\ \cos \frac{3\pi}{2}\right)$

Tuple Encoding

- Let $(\mathbf{A}, \mathbf{dom})$ be a tabular signature, $\mathbf{A} = A_1, \ldots, A_n$ and $\mathbf{dom} = dom_1, \ldots, dom_n$
- An *encoding scheme* Enc for $(\mathbf{A}, \mathbf{dom})$ associates with every column $i = 1, \ldots, n$:
  - A dimension $d_i \in \mathbb{N}$
  - A value encoding function $e_i : dom_i \to \mathbb{R}^{d_i}$
- A tuple $\mathbf{t}$ is encoded as a concatenation of its value embeddings:

$$Enc(\mathbf{t}) := e_1(\mathbf{t}[A_1]) \oplus \cdots \oplus e_n(\mathbf{t}[A_n]) \in \mathbb{R}^{d_1 + \cdots + d_n}$$

- We may also have another encoding layer that further processes the result
  - (e.g., lowering the dimension)

Encoding Scheme for a Relational Schema

An *encoding scheme* Enc for a relational schema $(\mathcal{R}, \text{sig}, \text{key}, \mathbf{FK})$ associates with each relation name R an encoding scheme Enc_R for $\text{sig}(R)$

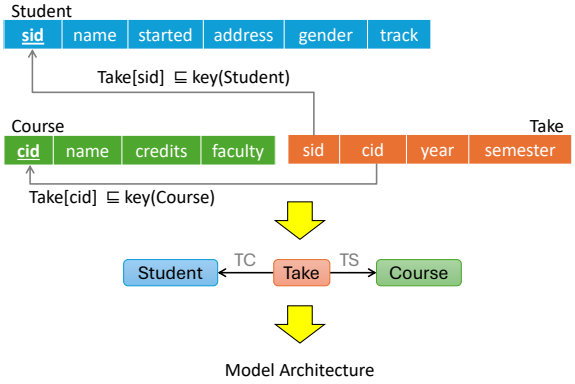
- Hence, the encoding of a tuple $\mathbf{t}$ in $D(R)$ is $Enc_R(\mathbf{t})$

The Common Conversion

- Let $\mathcal{S} = (\mathcal{R}, \text{sig}, \text{key}, \mathbf{FK})$ be a relational schema and:
  - D be a database over $\mathcal{S}$
  - Enc be an encoding scheme for $\mathcal{S}$
- The hetero-graph $\mathcal{G}_D = (L_V, L_E, V, E, \lambda, \mathbf{X})$ is defined as follows:
  - Every fact $R(\mathbf{t})$ becomes a node v with $\lambda(v) = R$ and $\mathbf{x}_v = Enc_R(\mathbf{t})$
  - For every FK $\Phi = R[\mathbf{C}] \sqsubseteq \text{key}(R')$ and facts $R(\mathbf{t})$ and $R'(\mathbf{t}')$ such that $R(\mathbf{t})$ references $R'(\mathbf{t}')$, we have the edge $(R(\mathbf{t}), \Phi, R'(\mathbf{t}'))$
  - In words, for every reference in the database, we add a corresponding edge labeled by the corresponding FK
- Note: if D and D' are databases of the same schema $\mathcal{S}$, then the hetero-graphs $\mathcal{G}_D$ and $\mathcal{G}_{D'}$ are type consistent, i.e., have the same hetero-schema

Why are Schemas Important to ML?

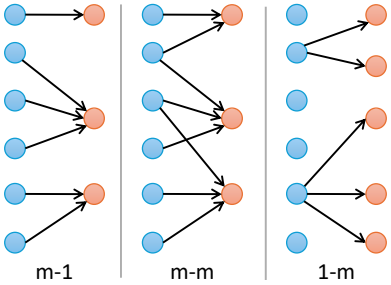
- A common approach to ML over databases is to train a model for a specific relational schema $\mathcal{S} = (\mathcal{R}, \text{sig}, \text{key}, \mathbf{FK})$
- The common methodology is to convert the database into a hetero-graph and train a model for the corresponding hetero-graph schema $(L_{\text{v}}, L_{\text{e}}, M, d)$



Mapping a Database Schema to a Hetero-Graph Schema

- Let:
  - $\mathcal{S} = (\mathcal{R}, \mathrm{sig}, \mathrm{key}, \mathbf{FK})$ be a relational schema
  - Enc be an encoding scheme for $\mathcal{S}$
- The schema of the hetero-graphs obtained from $\mathcal{S}$ is $(L_{\mathsf{v}}, L_{\mathsf{e}}, M, \delta)$ where:
  - $L_{\mathsf{v}} := \mathcal{R}$
  - $L_{\mathsf{e}} := \mathbf{FK}$
  - M maps every edge label (FK) $\Phi = R[\mathbf{C}] \sqsubseteq \mathrm{key}(S)$ into the pair (R, S)
    - That is, Φ connects an R-node to an S-node
  - δ maps every relation name $R \in \mathcal{R}$ into the output dimension of Enc_R

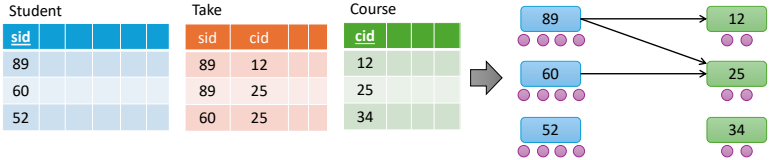
Many-to-One vs. Many-to-Many



- Importantly, FKs represent many-to-one relationships among nodes: a node can have **at most one outgoing edge** of a given type (FK)
- Hence, the out-degree of a node is bounded by the number of FKs from the node's relation
- This restriction implies that we construct only *certain types of graphs*
- Graph learning can be highly sensitive to the kind of conversion
- Hence, other conversions studied

Row2N/E

- Another conversion allows to designate some relationships with multiple FKs as converting directly into edges [Wang et al., 2024]
- Specifically, every pair of FKs from R represents an edge type in some direction

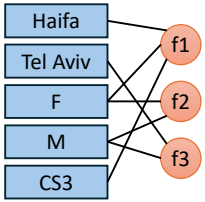


- Which direction? For common ML models, this is *not* important
- What about features of the connecting table?
  - These can be encoded as *edge features*

Cells as Nodes

Another example of a conversion: every *cell value* and every *fact* become a node, and each fact connects to its values [Cappuzzo et al., 2020]

	address	gender	track
f1	Haifa	F	CS3
f2	Haifa	M	CS4
f3	Tel-Aviv	F	SE



Conversion in the Course

Throughout the course, we will assume the Row2Node (FK-to-edge) conversion

- Which is most studied in the literature of deep learning over relational databases

Outline

1 Data Models
- Graphs
- Tables
- Relational Databases

2 Databases as Hetero-Graphs

3 Required Graph Concepts
- Neighborhood
- Matrix Representations of Graphs
- Weisfeiler-Leman Isomorphism Test

4 ML on Structured Data
- General ML
- Predictive Tasks
- Training

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Neighbor Notation

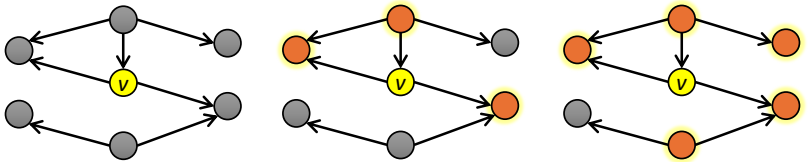
- Let G be a graph and v be a node of G
- We denote by:
  - $\text{Nodes}(G)$ the set of nodes of G
  - $N_G(v)$ the set of *neighbors* of v, that is, the nodes u of G that share an edge (in either direction) with v
    - Note: we follow the convention that $v \notin N_G(v)$
  - $N_G^{\text{in}}(v)$ the set of neighbors u of v via an *incoming* edge (u, v) into v
  - $N_G^{\text{out}}(v)$ the set of neighbors u of v via an *outgoing* edge (v, u) from v
- We usually remove the subscript G when G is clear from the context
  - E.g., $N(v)$ instead of $N_G(v)$

Node Degrees

Degrees are defined accordingly:

$$\deg_G(v) := |N_G(v)| \qquad \deg_G^{\mathsf{in}}(v) := |N_G^{\mathsf{in}}(v)| \qquad \deg_G^{\mathsf{out}}(v) := |N_G^{\mathsf{out}}(v)|$$

k-Hop Neighborhood



0-hop neighborhood 1-hop neighborhood 2-hop neighborhood

- Let $G = (V, E)$ be a graph and v be a node of G
- For $k \in \mathbb{N}$, the *k-hop neighborhood of v*, denoted $N_G^k(v)$, is the set of nodes in all undirected paths of length at most k, starting from v
  - That is, $N_G^k(v)$ consists of all nodes of distance at most k from v
- Note: v belongs to the k-hop neighborhood of v
- Note: literature often refers to the k-h.n. as the graph induced by $N_G^k(v)$

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Matrix Representations of Graphs

- It is very often convenient to represent a graph as a matrix
  - It allows for the extraction of interesting properties of the graph
  - Importantly, *spectral analysis* of graphs refers to the study of a graph via the spectrum—eigenvalues and eigenvectors—of matrix representations
  - It allows us to represent ML models on graphs as learnable numerical formulas over matrices (and tensors)
- We will encounter various matrices throughout the course; we define them here

The Adjacency Matrix

- Most importantly, the *adjacency matrix* of an undirected graph $G(V, E)$ with $V = \{v_1, \ldots, v_n\}$ is the matrix $\mathbf{A}$ with

$$\mathbf{A}_{i,j} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \in E \\ 0 & \text{otherwise} \end{cases}$$

  - The degree of a node is obtained by summing either its row or its column
- For a directed graph replace $\{v_i, v_j\}$ with (v_i, v_j)
  - The *in-degree* of a node is obtained by summing its column
  - The *out-degree* by summing its row

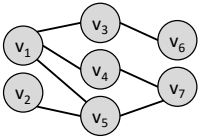
Degree and Laplacian Matrices

- Two other matrices commonly associated with an undirected graph $G(V, E)$ where $V = \{v_1, \ldots, v_n\}$: the *degree matrix* and the *graph Laplacian*
  - Various variants exist for directed graphs
- The degree matrix $\mathbf{D} \in \mathbb{R}^{n \times n}$ is a diagonal matrix:

$$\mathbf{D}_{i,j} := \begin{cases} \deg(v_i) & \text{if } i = j\,; \\ 0 & \text{otherwise.} \end{cases}$$

  - We can similarly define the *out-degree* matrix and the *in-degree* matrix
- The graph Laplacian is defined by $\mathbf{L} = \mathbf{D} - \mathbf{A}$
- When G is undirected, each of $\mathbf{A}$, $\mathbf{D}$ and $\mathbf{L}$ are *symmetric*

Example



A

0	0	1	1	1	0	0
0	0	0	0	1	0	0
1	0	0	0	0	1	0
1	0	0	0	0	0	1
1	1	0	0	0	0	1
0	0	1	0	0	0	0
0	0	0	1	1	0	0

D

3	0	0	0	0	0	0
0	1	0	0	0	0	0
0	0	2	0	0	0	0
0	0	0	2	0	0	0
0	0	0	0	3	0	0
0	0	0	0	0	1	0
0	0	0	0	0	0	2

L

3	0	-1	-1	-1	0	0
0	1	0	0	-1	0	0
-1	0	2	0	0	-1	0
-1	0	0	2	0	0	-1
-1	-1	0	0	3	0	-1
0	0	-1	0	0	1	0
0	0	0	-1	-1	0	2

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Graph Isomorphism

- $G = (V, E)$ and $G' = (V', E')$ are *isomorphic* if there is a bijection $\varphi : V \to V'$ such that for all $v, u \in V$ it holds that $(v, u) \in E$ if and only if $(\varphi(v), \varphi(u)) \in E'$
  - In the case of an undirected graph: $\{v, u\} \in E$ if and only if $\{\varphi(v), \varphi(u)\} \in E'$
- Note: two graphs G_1 and G_2 with adjacency matrices $\mathbf{A}_1$ and $\mathbf{A}_2$ are isomorphic iff there is a *permutation matrix* $\mathbf{P}$ such that $\mathbf{A_1} = \mathbf{P}\mathbf{A_2}\mathbf{P}^\top$
  - A *permutation matrix* is a matrix with precisely one 1-cell in each row and in each column, and 0 everywhere else
  - Idea: first multiplication permutes the rows, second permutes the columns
- Graph isomorphism is not known to be solvable in polynomial time
  - It is in NP, but conventionally believed to be easier than NP-hard problems
  - In contrast, it is NP-complete to decide whether a graph G is *isomorphic to a subgraph* of another graph G'

Why is Graph Isomorphism Important to us?

- Consider a graph G and two nodes v and u of G
- If $N^k(v)$ and $N^k(u)$ induce isomorphic subgraphs (mapping v to u), then *any algorithm* that looks only at k-hops cannot distinguish between v and u
  - No matter how "deep" or "full-of-parameters" the algorithm is
- As we will see in the course, this will also be true for an important class of algorithms (GNNs), even if we replace isomorphism with a *weaker* notion that is easy to compute (in contrast to isomorphism), namely **WL**
  - An idea from the 1960s, nowadays found very useful in graph ML design & analysis
- This will shed light on the expressiveness and limitations of the families of functions we learn over graphs

Why do we Care about WL?

- A tool for feature generation in graph ML
- A tool for analyzing the expressive power of GNNs
- A concept that inspired new GNN architectures
  - Graph Isomorphism Networks [Xu et al., 2019]
  - Deep Homomorphism Networks [Maehara and NT, 2024]

Class Discussion

Suggest efficient ways of detecting that two given graphs G_1 and G_2 are non-isomorphic

Weisfeiler-Leman (WL) Isomorphism Test (Intuition)

- Suppose that we wish to quickly prove that G_1 and G_2 are **not** isomorphic
  - If the node sets of G_1 and G_2 have different sizes, we're done
  - Otherwise, group the nodes by their degree; if any two corresponding groups have different sizes, then we're done
  - Otherwise, further group nodes by the collection of the degrees of their neighbors
  - ...




- The Weisfeiler-Leman isomorphism test [Weisfeiler and Leman, 1968] applies this process repeatedly until the classification cannot be further refined
- In each round, nodes v are being "colored" by a natural number $\kappa(v) \in \mathbb{N}$ that represents the group description (e.g., "all nodes of degree 3")
- Referred to as *WL*; also *1-WL* (k-WL colors k-sequences of nodes)

Undirected WL

- WL is based on a hash (coloring) function $\textsc{Hash}$ that takes as input a node color $c \in \mathbb{N}$ and a bag B of neighbor colors, and returns a color $\textsc{Hash}(c, B) \in \mathbb{N}$
  - By a *bag* we mean a set where each element is assigned a multiplicity
- Crucially, $\textsc{Hash}(c, B)$ is assumed to be *injective*:

$$\textsc{Hash}(c, B) = \textsc{Hash}(c', B') \text{ iff } c = c' \text{ and } B = B'$$

- Two colorings $\kappa : V \to \mathbb{N}$ and $\kappa' : V \to \mathbb{N}$ for a graph $G = (V, U)$ are *equivalent* if they partition the node set to the same groups; that is:

$$\text{For all } v, u \in V \text{ we have } \kappa(v) = \kappa(u) \text{ iff } \kappa'(v) = \kappa'(u)$$

The Undirected WL(V, E) Procedure

1. Initialize $\kappa_0(v) = 0$ for all $v \in V$
2. Having constructed κ_i, define κ_{i+1} as follows for all nodes v:

$$\kappa_{i+1}(v) = \textsc{Hash}\big(\kappa_i(v), \{\!\{\kappa_i(u) \mid u \in N(v)\}\!\}\big)$$

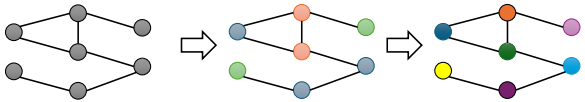
3. Stop when κ_{i+1} and κ_i are equivalent

Note: by $\{\!\{\kappa_i(u) \mid u \in N(v)\}\!\}$ we mean the *bag* (rather than *set*) of colors $\kappa_i(u)$

Class Discussion

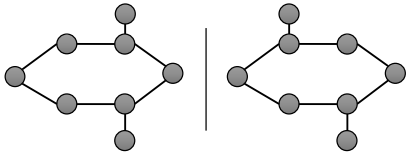


Examples

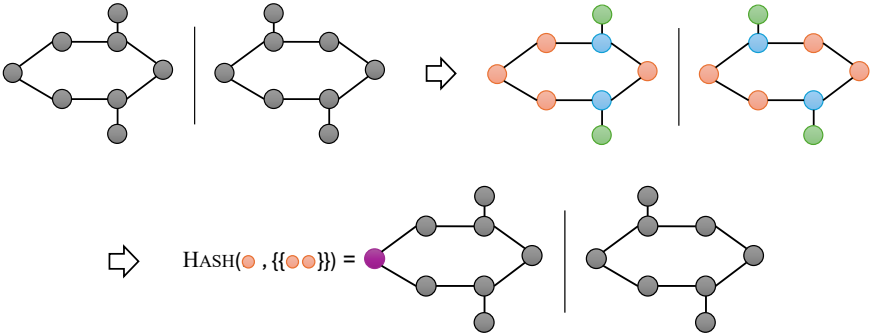


Class Discussion

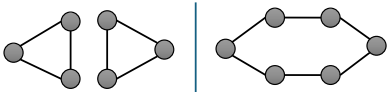
Can you use 1-WL to tell that the following are non-isomorphic?



WL for Isomorphism Testing



Non-isomorphic, but WL cannot tell:



Understanding WL I

Theorem

If κ_{i+1} and κ_i are equivalent and we continue WL indefinitely, then κ_j and κ_i are equivalent for every $j \geq i$

- Hence, the stopping criterion terminates the process when reaching a stable state
- **Proof idea:** The state at step $i+1$ is "equivalent" to the state at step i, so HASH will be called with similar inputs, up to recoloring, and will thus produce similar outputs, hence κ_{i+2} is equivalent to κ_{i+1}, and so on (induction)

Understanding WL II

Lemma

For all $i \geq 0$ and nodes $v, u \in V$, if $\kappa_i(v) \neq \kappa_i(u)$, then $\kappa_{i+1}(v) \neq \kappa_{i+1}(u)$.

- Hence, the grouping induced by κ_{i+1} is a refinement (i.e., obtained by splitting groups) of the grouping κ_i
- **Proof idea:** From the ith step on, $\textsc{Hash}(c, B)$ will be called with two different colors c, and hence produce two different new colors c'

Understanding WL III

Theorem

WL terminates after at most $|V|$ steps

- Hence, we can execute WL efficiently
- **Proof idea:** Due to the previous lemma, every step strictly refines the previous grouping, and we can have at most $|V|$ groups

Understanding WL IV

Theorem

Let φ be an isomorphism between the graphs $G = (V, E)$ and $G' = (V', E')$. If we run WL in parallel over G and G', then $\kappa_i(v) = \kappa_i(\varphi(v))$ for all $i \geq 0$.

- Hence, a necessary condition for G and G' to be isomorphic is that each τ_i produces the same number of occurrences of every color
- **Proof idea:** Straightforward induction on the step number

Directed and Labeled WL

- Directed: (e.g., [Rossi et al., 2023])

$$\kappa_{i+1}(v) = \textsc{Hash}\Big(\kappa_i(v)\,,\ \{\!\{\kappa_i(u) \mid u \in N^{\mathsf{in}}(v)\}\!\},\{\!\{\kappa_i(u) \mid u \in N^{\mathsf{out}}(v)\}\!\}\Big)$$

- Labeled: $\kappa_0(v) = \lambda(v)$
  - Identifying labels as natural numbers
- Featured & labeled (heterogeneous): $\kappa_0(v) = \textsc{Hash}'(\lambda(v), \mathbf{x}_v)$
  - For some perfect hashing function $\textsc{Hash}'$
  - We need to assume that $\mathbf{x}_v$ comes from a *countable domain* (e.g., $\mathbb{Z}^d$) for such $\textsc{Hash}'$ to exist

That's it! The lemma and theorems continue to hold

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Core ML Task

- We have a space $\mathcal{X}$ of *examples* (*instances*), a *target space* $\mathcal{Y}$
- We have an unknown function $f : \mathcal{X} \to \mathcal{Y}$ that we wish to mimic
- For that, we are getting a set S of labeled examples (x_i, y_i) where $x_i \in \mathcal{X}$ and $y_i \in \mathcal{Y}$, representing values of f on different inputs ($y_i = f(x_i)$)
- We have a hypothesis space $\mathbf{H}$ of functions (models) $h : \mathcal{X} \to \mathcal{Y}$
  - $\mathbf{H}$ does not necessarily include f
- Our goal is to find a hypothesis $h \in \mathbf{H}$ that is closest to f
- To this end, we assume that the labeled examples of S are chosen randomly
  - (e.g., i.i.d.)

Common Types of ML Tasks

- ML tasks traditionally classified into different types, based on the target space $\mathcal{Y}$
  - Binary classification: $\mathcal{Y} = \{0, 1\}$
  - Multi-class classification: $\mathcal{Y} = \{0, \ldots, K-1\}$ for $K \geq 2$
  - Multi-label classification: $\mathcal{Y} = \mathcal{P}(\{0, \ldots, K-1\})$ – each x mapped to a *set* of labels
  - Regression: $\mathcal{Y} = \mathbb{R}$
  - Multivariate regression: $\mathcal{Y} = \mathbb{R}^k$
- Commonly, a binary classification model h is learned as a regression model h' (e.g., a probability function $h' : \mathcal{X} \to [0, 1]$), followed by thresholding:

$$h(x) := \begin{cases} 0 & \text{if } h'(x) < \tau; \\ 1 & \text{otherwise.} \end{cases}$$

- We will also see multi-class classification via multivariate regression

Training

- *Training* is the process of computing the candidate $h \in \mathbf{H}$, given S
- Conventionally, this is done by adopting a *loss function* $\text{Loss}(S, h)$ over the set

$$\{(x_i, y_i, h(x_i)) \mid (x_i, y_i) \in S\},$$

  stating how far off h is
- We then apply an algorithm for finding h that minimizes $\text{Loss}(S, h)$
- For example $\text{Loss}(S, h) := \sum_{(x_i, y_i) \in S} \|y_i - h(x_i)\|$ when $(\mathcal{Y}, \|\cdot\|)$ is a norm space

Parametric Models

- Most prominent examples of ML are cases where $\mathbf{H}$ is of the form $\{h_{\boldsymbol{\theta}} \mid \boldsymbol{\theta} \in \mathbb{R}^k\}$ where $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_q)$ is a sequence of *numerical parameters*
  - Also called *weights* or *coefficients*
  - Same number q of parameters for every model in $\mathbf{H}$
- Hence, the goal is to compute the best $\boldsymbol{\theta}$
- But there are other models, called *non-parametric*, where different hypotheses differ in their entire form, not (just) the parameters
  - Most importantly (to us), *decision trees*
  - Also, *k-Nearest-Neighbor*

Regularization

- A typical addition is a component that introduces a bias toward "preferred" models; specifically, *regularization* penalizes h for being complex
- Rationale: complex functions might *overfit*, that is, hang on to nuances of the examples
  - Simplicity forces the choice of h to grasp the big picture and better generalize

Common Loss Functions for Numerical Parametric Prediction

- Denote $\hat{y}_i = h(x_i)$, the predicted value (aiming to be close to y_i)
- $\text{Loss}(S, h_{\boldsymbol{\theta}}) := \text{Err}(S, h_{\boldsymbol{\theta}}) + \lambda \cdot \text{Reg}(h_{\boldsymbol{\theta}})$ (where λ tunes the level of regularization)

Error functions: (for regression)

- Mean Squared Error: $\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$
- Mean Absolute Error: $\frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$
- Huber Loss: $\frac{1}{n}\sum_{i=1}^{n} L(y_i - \hat{y}_i)$ where

$$L(r) = \begin{cases} \frac{1}{2}r^2 & \text{if } |r| < \delta; \\ \delta(|r| - \frac{1}{2}\delta) & \text{otherwise.} \end{cases}$$

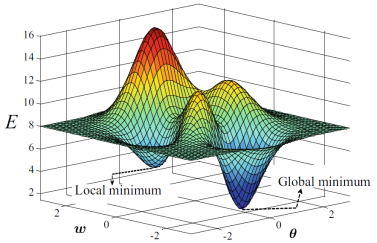
  - Combination between MSE (on small values) and MAE (on large values)

- Cross Entropy: (for binary/multiclass) $-\frac{1}{n}\sum_{i=1}^{n}\sum_{c} y_{i,c}\log(\hat{y}_{i,c})$
  - $y_{i,c}$ indicates whether y_i is labeled c ($y_{i,c} = 1$) or not ($y_{i,c} = 0$)
  - Assumes $0 < \hat{y}_{i,c}$

Regularization terms:

- L1 (Lasso regression): $\sum_{\theta_i \in \boldsymbol{\theta}} |\theta_i|$
- L2 (Ridge regression): $\sum_{\theta_i \in \boldsymbol{\theta}} \theta_i^2$

Fitting via Optimization

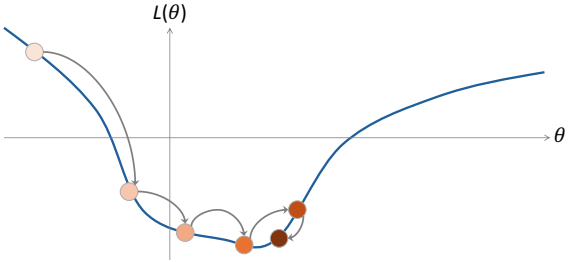


- Importantly, we translate ML into an algorithmic optimization problem
- Some traditional ML algorithms solve it exactly, e.g., *Support Vector Machines*
- Yet, for complex models with many parameters, training on large volumes of data, a guaranteed optimum is typically computationally intractable, and impractical

Gradient-Based Optimization

Hence, we use heuristic techniques that are successful in practice

- And oftentimes prove their quality analytically under assumptions
- The most common techniques are *gradient-based* methods



Gradient Descent

- Denote $L(\boldsymbol{\theta}) := \text{Loss}(S, h_{\boldsymbol{\theta}})$
- Assume that L is differentiable, and we can compute the gradient:

$$\nabla L(\boldsymbol{\theta}) := \left(\frac{\partial L}{\partial \theta_1}(\boldsymbol{\theta}), \ldots, \frac{\partial L}{\partial \theta_q}(\boldsymbol{\theta}) \right)$$

- *Gradient descent* (GD) iteratively updates $\boldsymbol{\theta}$ via steps of the form

$$\boldsymbol{\theta} := \boldsymbol{\theta} - \eta \cdot \nabla L(\boldsymbol{\theta})$$

where $\eta \in \mathbb{R}_+$ is a positive *step size* (or *learning rate*)

- Until a stopping condition is reached

Class Discussion

What are the standard difficulties with vanilla GD?

Problems with Vanilla GD

- **Costly gradient computation**: We may have too many examples to compute the gradient on the whole training set S on each step
  - Optimization is often done in batch-oriented hardware (e.g., GPUs, to be discussed later in the course), and $\nabla L(\boldsymbol{\theta})$ may be too big to fit when S is large
- **Anomalous convergence**: Which learning rate η to use?
  - With a large η, GD can get way off due to crude jumps
  - With a small η, GD can get stuck in local minima and generally converge too slowly
  - The fundamental issue is that the learning rate η is fixed, no matter where we are in the learning process

Stochastic Variants

- Efficiency & randomness introduced by the *stochastic* variants of GD
- *Stochastic GD* (SGD): pick a random sample $(x_i, y_i) \in S$, typically without replacement, and update only based on this example!

$$\boldsymbol{\theta} := \boldsymbol{\theta} - \eta \cdot \nabla L_i(\boldsymbol{\theta}) \quad \text{where} \quad L_i(\boldsymbol{\theta}) := \text{Loss}(x_i, y_i, h_{\boldsymbol{\theta}})$$

- *Mini-batch SGD*: pick a random *mini-batch* $B \subseteq S$, without replacement, and update only based on this batch:

$$\boldsymbol{\theta} := \boldsymbol{\theta} - \eta \cdot \nabla L_B(\boldsymbol{\theta}) \quad \text{where} \quad L_B(\boldsymbol{\theta}) := \text{Loss}(B, h_{\boldsymbol{\theta}})$$

  - Every *epoch* is a round that completes all batches (and then we can go again)

Adam Optimizer [Kingma and Ba, 2015]

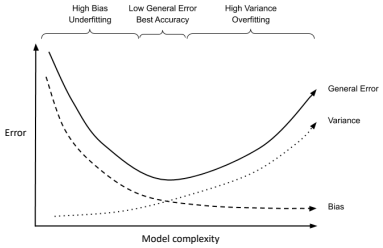
- State-of-the-art form of GD with *adaptive learning rates* that combines known approaches of momentum and RMSProp (Root Mean Square Propagation)
- Maintains a balance between efficient convergence and stability by incorporating an exponentially weighted *moving average of past gradients*
- Past gradients diminish their impact in time (forgotten)

Specification of the Adam Optimizer

- Start with $m_0 = v_0 = g_0 = 0$
- Let t denote the update number, and g_t the gradient of $\boldsymbol{\theta}$ at time t
- *1st & 2nd moments* (for hyper-parameters β_1 and β_2 in $(0, 1)$):
  - $m_t = g_t \cdot (1 - \beta_1) + m_{t-1} \cdot \beta_1$ (mean estimate)
  - $v_t = g_t^2 \cdot (1 - \beta_2) + v_{t-1} \cdot \beta_2$ (variance estimate)
- *Bias correction:* $\hat{m}_t = \frac{m_t}{1-\beta_1^t}$; $\hat{v}_t = \frac{v_t}{1-\beta_2^t}$
  - Diminishing amplification
- *Parameter update:* $\boldsymbol{\theta}_t = \boldsymbol{\theta}_{t-1} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$ for hyper-parameters α and ϵ
- Common defaults: $\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$

Fitting Risks

- Standard challenges in ML:
  - Under-fitting: Fail to fit the training examples, bad performance during training
  - Over-fitting: Fit well the training set, but generalize poorly to instances outside
- These can be viewed via the *bias-variance tradeoff* of our setting—hypothesis class $\mathbf{H}$ and learning algorithm
  - Bias: How biased are we towards wrong/simplistic assumptions (e.g., too simple hypotheses)?
    - *High bias means that we can poorly fit our samples*
  - Variance: How sensitive are we to the specifics of the samples (e.g., too complex hypotheses)?
    - *High variance means that we can get highly different things on different sample sets*

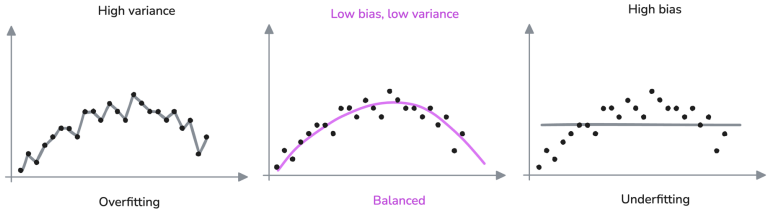


https://www.ml-science.com/

Class Discussion

How do we manage the bias-variance tradeoff?

Managing the Bias-Variance Tradeoff



- To handle under-fitting, we use more complex models (e.g., additional layers with more parameters), apply additional epochs, use better SGD update strategies
- To handle over-fitting, we use better regularization, and control our ML development using proper *data splitting*

[2] Figure from https://briefgenai.com/blog/fundamentals/bias-variance-under-over-fitting

Data Splitting

- Conventional ML splits the data into three parts:

  1. Train data **2.** Validation data **3.** Test data

- The main training process (e.g., SGD) is applied to the **training data**
- **Validation data** is used for *hyper-parameter tuning* (e.g., GD step size, number of layers, regularization weights), *model selection* (in case we train multiple models and choose the best), and *performance monitoring* (e.g., estimating how well we are during SGD) ; not "processed" — just validates
- The **test data** is used for reporting the predicted quality of the model
- Another common split is a holdout data that is allowed to be seen only at the very end (and, once seen, becomes invalid)
  - This is to avoid overfitting of the whole ML development (by the programmers)

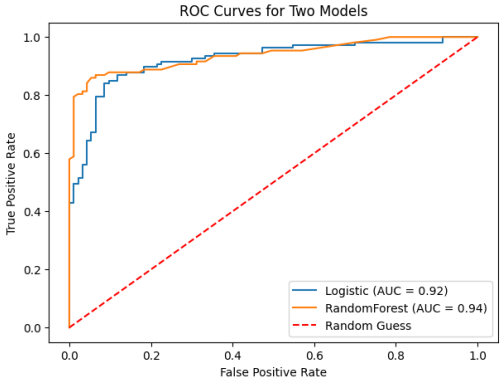
Quality Evaluation of a Learned Model

- How do we measure the quality of a model over the test set?
- We can use the error component of the loss function (or some other loss function)
- Important case: threshold-based **binary classifier**
- We would like to measure not just the result (with a specific threshold), but also the ability of the score to support good classification
- A popular measure is the *area under the curve of the receiver operating characteristic* (**ROC-AUC**)
- ROC-AUC is the probability that a randomly chosen "yes" (1) instance is ranked higher than a randomly chosen "no" (0) instance
  - For example, ROC-AUC of a *random score* is 0.5
- The name comes from the *ROC curve*:

Class Discussion

Why “area under the curve”? Which curve?

Visual Interpretation of ROC-AUC (and why *AUC*?)



https://www.geeksforgeeks.org/machine-learning/auc-roc-curve/

- Sort the 0-instances by decreasing score on the x-axis: $x_1, \ldots, x_n$
- Each x_i defines a *false positive rate*: the portion of 0-instances scored at least as x_i, namely i/n
- For each x_i, mark its y-value as the ratio of *true positives* (i.e., % of 1-instances scored above x_i) — called the ROC curve
- Our probability is the *area under the curve*, i.e., the probability that a random point is placed under the curve

Outline

1. Data Models
   - Graphs
   - Tables
   - Relational Databases
2. Databases as Hetero-Graphs
3. Required Graph Concepts
   - Neighborhood
   - Matrix Representations of Graphs
   - Weisfeiler-Leman Isomorphism Test
4. ML on Structured Data
   - General ML
   - Predictive Tasks
   - Training

Predictive Tasks on Graphs: Generic Form

- A general class of functions we learn over graphs have the shape

$$F : \mathbf{G} \times \mathbf{V}^k \to \mathbb{R}^\ell$$

  where:
  - $\mathbf{G}$ is a class of graphs
  - $\mathbf{V}$ is the domain of graph nodes
  - $F(G, \mathbf{v})$ is defined only when $\mathbf{v}$ is a sequence of k nodes from G
- In the case of featured graphs, we require that all graphs in $\mathbf{G}$ have the same feature dimension d
- In the case of hetero-graphs, we require that all graphs in $\mathbf{G}$ are type consistent
- Next, we see examples

Predictive Tasks on Graphs: Node Tasks

- Node classification: $\mathbf{G} \times \mathbf{V} \to \{0, 1\}$
  - For example, student's graduation within four years (students represented as nodes)
  - This variant is *binary classification*; *multiclass classification* has the form $\mathbf{G} \times \mathbf{V} \to \{0, \ldots, K-1\}$
- Node regression: $\mathbf{G} \times \mathbf{V} \to \mathbb{R}$
  - For example, student's average grade (or #credit points) by the end of the year
- Node embedding: $\mathbf{G} \times \mathbf{V} \to \mathbb{R}^k$
  - For example, in *protein folding* we have $\mathbf{G} \times \mathbf{V} \to \mathbb{R}^3$ where each node represents an amino acid, and we wish to predict its position in a folded state
  - Often used to facilitate link prediction
- Node clustering: $\mathbf{G} \times \mathbf{V} \to \mathbb{N}$
  - $F(G, v) = i$ means that v is assigned to cluster number i
  - Also known as *community detection*

Predictive Tasks on Graphs: beyond Nodes

- Link prediction: $\mathbf{G} \times \mathbf{V}^2 \to \{0, 1\}$
  - For example, predict which student will take which course
- Graph-level tasks: classification: $\mathbf{G} \to \{0, 1\}$, regression $\mathbf{G} \to \mathbb{R}$, graph embedding $\mathbf{G} \to \mathbb{R}^k$, graph clustering $\mathbf{G} \to \mathbb{N}$
  - Example: Is a molecule toxic (classification)? What is the level of toxicity (regression)?
  - More examples: Embed molecules on a screen so that similar ones are close to each other, and vice versa (embedding); group molecules by their behavior (clustering)
- . . . and other variations, e.g., link regression, predict multi-ary relationships, etc.

Predictive Tasks on Tables

- Let $(\mathbf{A}, \mathbf{dom})$ be a tabular signature, $\mathbf{A} = A_1, \ldots, A_n$, $\mathbf{dom} = dom_1, \ldots, dom_n$
- Row regression/classification: $F : dom_1 \times \cdots \times dom_n \to \mathbb{R}^\ell$ (classic ML)
- Column prediction: Given a table r with the column A_i missing, complete the missing column
  - The domain dom_i of A_i can be Boolean (binary classification), categorical (multiclass classification), or numerical (regression)
- Missing data imputation: Given a table r' with several cells missing, guess the completion r of r'
- Data cleaning: Given a table r' with errors in some cells, guess which cells have wrong values, and guess a correction r of r'
- Entity resolution: Given a table r, guess rows refer to the same real-world entity (a.k.a. *de-duplication*)

Predictive Tasks on Databases

- Let $\mathcal{S} = (\mathcal{R}, \text{sig}, \text{key}, \mathbf{FK})$ be a database schema
- Denote by $\mathbf{D}_\mathcal{S}$ the set of all databases over $\mathcal{S}$
- Column prediction / missing data imputation / cleaning: Same as table tasks on a designated *target relation* $T \in \mathcal{R}$, given the entire database D (not just $D(T)$)
- Entity linking/matching: given a database D and two relation names $R, S \in \mathcal{R}$, guess which tuples of R are linked to which tuples of S
  - $F : \mathbf{D}_\mathcal{S} \times dom(R) \times dom(S) \to \{0, 1\}$
  - E.g., which Customer rows will purchase which Product rows
- Relational forecasting: database changes over time—D_t is the database at time t; fix a numerical query $Q : \mathbf{D}_\mathcal{S} \to \mathbb{R}$; predict the answer $Q(D_t)$ at a future time t
  - For example, predict the October 2026 #sales for customers from Berlin

A Note about Invariance of Models

- An important property of the class of models we learn is *invariance* to aspects of the database that should not matter to us
  - And are expected to be changed without implication on the model
- Examples:
  - Order of columns in each relation
  - Order of the tuples
  - Opaque keys that we use as artificial identifiers
  - *Anything else?*
    - For example, one might want invariance to the *attribute names*, another may not
- We need to be careful in how we define *invariance*; for example, column reordering will change the order of encoders—might break "legitimate" invariant models

Schema-Level Tasks

- *Schema-level tasks* predict concepts that apply to the schema and not a specific database, but can be learned using specific databases
- Schema matching: Given two schemas $\mathcal{S}$ and $\mathcal{S}'$, determine which pairs of attributes A and A' from the two refer to the same information (e.g., a person's name, a company's location)
- Query learning: Given answers to a query (e.g., SQL), guess what the query is
- Text-to-SQL: Given a schema $\mathcal{S}$, translate a textual description of a query into an SQL query

Schema-level tasks are not covered in the course!

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Transductive vs. Inductive Learning in Graphs

- Consider a node-classification task on graphs
- We distinguish between two types of *learning problems*:
  - Inductive: What we considered so far, where we learn a model $F : \mathbf{G} \times \mathbf{V} \to \{0, 1\}$ that can be applied to unseen graphs
    - For example, we learn the *weights* of a general graph model (e.g., a GNN)
  - Transductive: We have a fixed graph $G = (V, E)$, treated as our *context*, for both training and evaluation time—train on labeled nodes, evaluate on the rest
    - This means that we actually wish to learn a function $F_G : V \to \{0, 1\}$
    - For example, we classify based on *actual node embedding* that we learn for G, and know nothing about new nodes
- Importantly, we can apply an **inductive** learning method on a **transductive** setting by providing it with the same graph G
- Same applies to all other graph-learning tasks (e.g., link prediction)

3 Levels of Relational Learning

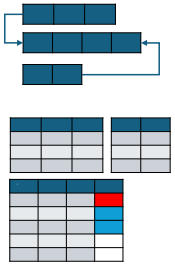
When learning over relational databases (or hetero-graphs), we can consider *three modes of generalization* (considering row-level tasks):

1. Transductive learning: Fixed schema, fixed database, unknown tuple label
   - Example: data imputation
2. Inductive learning: Fixed schema, unknown database, unknown tuple label
   - Example: column prediction
3. Foundation models: Unknown schema, unknown database, unknown tuple label

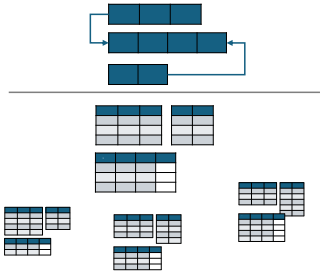
In the course, we will focus mainly on the *inductive setting*, and toward the end discuss *foundation models*

Generalization Modes: Transductive, Inductive, Foundation

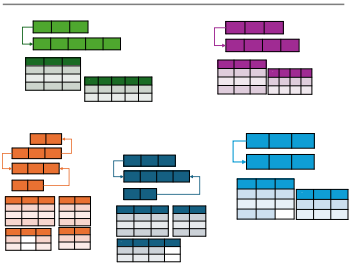
Transductive model:
Known schema, known task, known DB



Inductive model:
Known schema, known task, unknown DB



Foundation model:
Unknown schema, unknown task, unknown DB



Training an Inductive Model in Transductive Conditions

- Consider a learning task on structured data, where we wish to learn a function $f(D, e)$ on an input instance that consists of a *data structure* D and an *entity* e
  - D is a graph or a database
  - e is a node, a row, a fact, a sequence of nodes, etc.
- In an inductive ML setting, we need to train on many examples (D_i, e_i, y_i)
  - This is the case, for instance, when D represents a molecule
- In practice, we often train in a *transductive setting* on a *single* D with many entities, that is, examples (D, e_i, y_i) that share the same D
  - But we still want an inductive model, since the structure D changes in time (e.g., new customers, etc.), and we want it to keep being applicable

Substructures in Mini-Batching

- When D is large, training steps on (D, e_i, y_i) cannot consume it fully
- To handle large structures, mini-batching often involves sampling from D a small sub-structure D_B
  - Hence, SGD is applied to the (D_B, e_i, y_i) of the mini-batch B
- For example, on graph structures, we can sample walks from *k-hop neighborhoods* of sample entities e_i [Hamilton et al., 2017], or take clusters [Chiang et al., 2019]

Conclusion: We commonly learn an inductive model on one big structure by translating the setup into a classic inductive setting with many small structures

Timestamp-Based Training

- When we learn to predict a future event, it is convenient to use timestamps for training
- Specifically, we assume that each fact f of a database D is associated with a timestamp $ts(f)$ that represents the creation time of the the fact
- Timestamps are then used in multiple ways:
  1. *Data splitting*: training data is the oldest, validation data is newer, and test data is the latest
  2. *Structure sampling* for mini-batches may avoid using facts that are newer than the tested target node
  3. The timestamp itself is used as an additional *attribute* of the fact and, consequently, encoded into the feature vector

Key Takeaways from Module 1

- We focus on 3 types of data structures: tables, graphs, and relational databases
- We use a natural translation of a relational database into a hetero-graph
- WL is a generic graph technique for gathering graph information about each node; necessary but insufficient condition for isomorphism
- Many types of ML tasks on each of the three data types
- Generalization modes: transductive (one database), inductive (one schema, input database), foundation models (input schema, input database)

The End

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